Student HTTP Web API Service

DATA2410 - Assignment #3

OsloMet - Faculty of Technology, Art and Design (TKD)

Date: 04.05.2026

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Contents

1 Overview 2

2 Setup and Run 2

3 Testing in Scalar 3

3.1 Calculate Student Grades 3

3.2 Calculate Student Report 3

3.3 Clear the database (optional) 4

4 Results 4

1 Overview

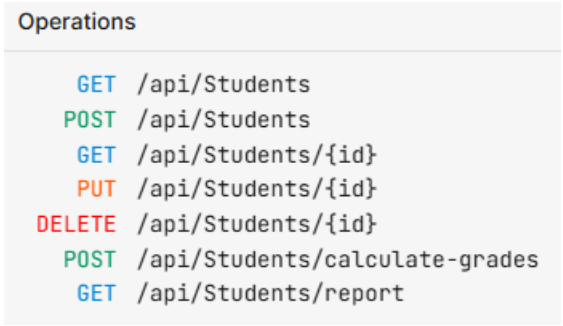
- Goal: Build two endpoints in an ASP.NET Core Web API with SQL Server (StudentsDb).
- Endpoints:
    - POST /api/Students/calculate-grades: compute and save A/B/C/D from Marks, return updated students.
    - GET /api/Students/report: per-course totals, average marks (2 decimals), and grade counts.
- CRUD was already there; added DELETE.

2 Setup and Run

- Create database StudentsDb in SSMS.
- appsettings.json connection: Server=localhost;Database=StudentsDb;Trusted_Connection=True;TrustServerCertificate=True;

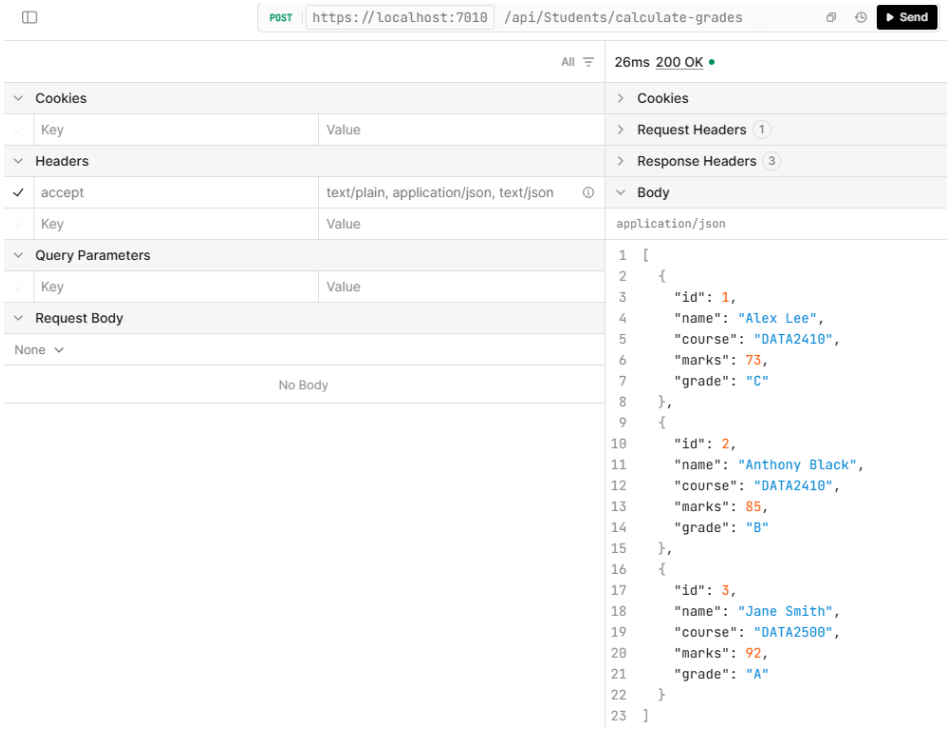


- Run:
    - dotnet restore .\data2410-api-v1.csproj
    - dotnet run --project .\data2410-api-v1.csproj
- Open docs (Scalar): https://localhost:<port>/scalar/v1



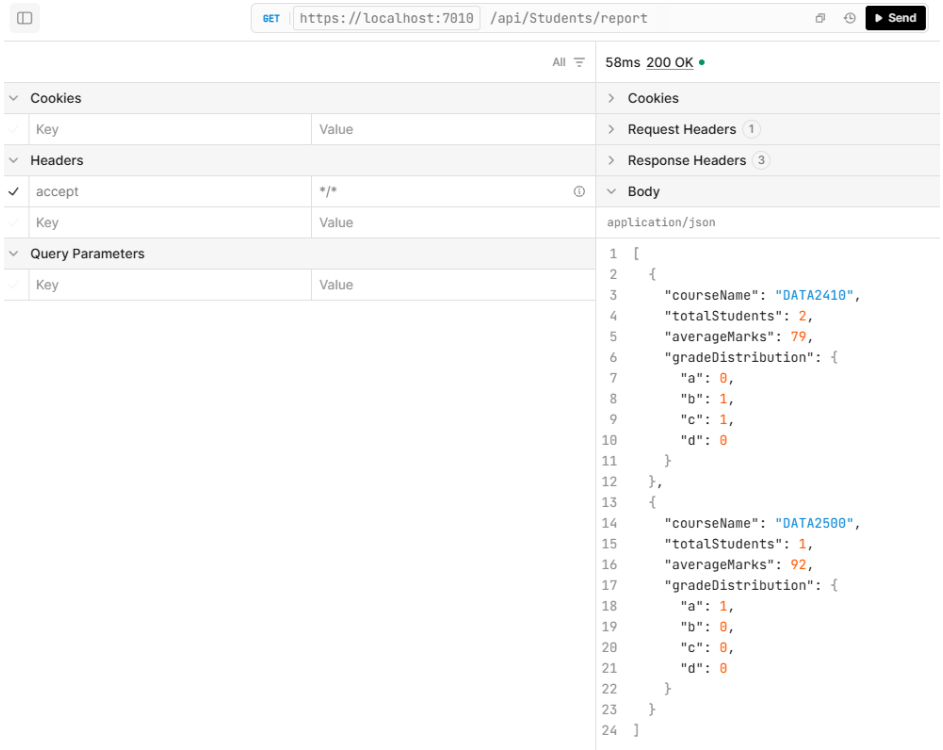
3 Testing in Scalar

3.1 Calculate Student Grades



POST /api/Students/calculate-grades

3.2 Calculate Student Report



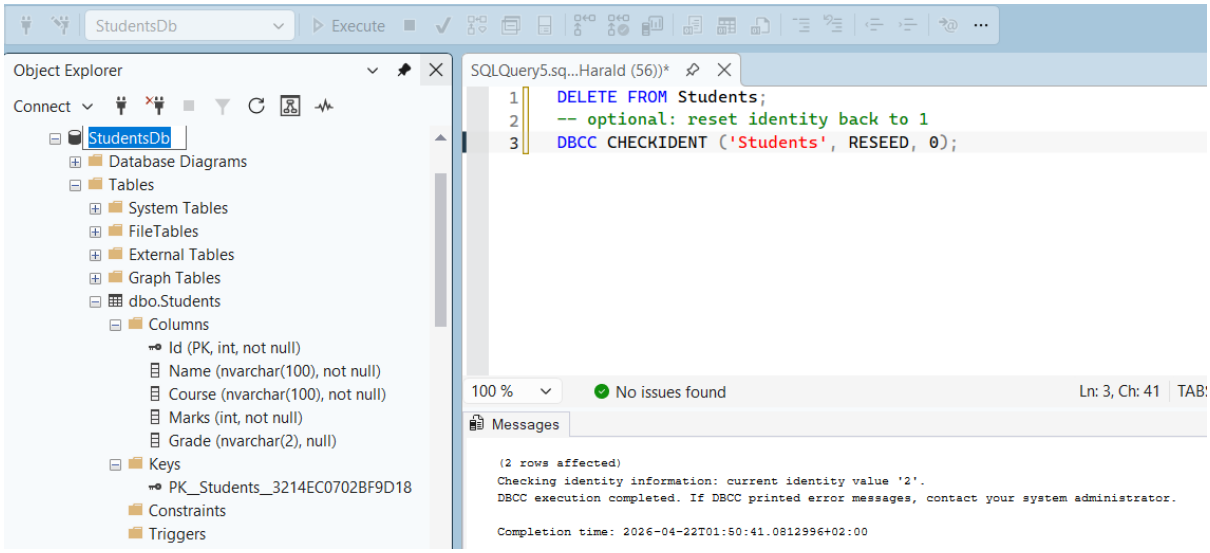
GET /api/Students/report

3.3 Clear the database (optional)

1. SSMS → connect to StudentsDb → New Query
2. Run:

   DELETE FROM Students;

   DBCC CHECKIDENT ('Students', RESEED, 0);
3. Verify in Scalar: GET /api/Students → should return [0]



4 Results

- The calculate-grades endpoint saves grades correctly based on the rules:
  - A ≥ 90, B = 80–89, C = 60–79, D < 60.
- The report endpoint shows the correct numbers per course:
  - totalStudents
  - averageMarks (2 decimals)
  - gradeDistribution (A/B/C/D)
- Example (from the latest test data):

```
  "courseName": "DATA2410",
  "totalStudents": 2,
  "averageMarks": 79,
  "gradeDistribution": {
    "a": 0,
    "b": 1,
    "c": 1,
    "d": 0
```

```
  "courseName": "DATA2500",
  "totalStudents": 1,
  "averageMarks": 92,
  "gradeDistribution": {
    "a": 1,
    "b": 0,
    "c": 0,
    "d": 0
```